

Problem

A parallel RLC circuit has $R = 100\text{ k}\Omega$, $L = 100\text{ mH}$, and a $C = 10\text{ }\mu\text{F}$. Determine the value of Q , the resonant frequency, and the bandwidth. If $R = 200\text{ k}\Omega$, how does that affect the values of Q , resonant frequency, and the bandwidth?

Step-by-step solution

Step 1 of 6

Calculate the resonant frequency.

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Substitute 100×10^{-3} for L and 10×10^{-6} for C .

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{LC}} \\ &= \frac{1}{\sqrt{100 \times 10^{-3} \times 10 \times 10^{-6}}} \\ &= 1000 \\ &= 1\text{ krad/s}\end{aligned}$$

Therefore, the resonant frequency, ω_0 is 1 krad/s.

Step 2 of 6

Write the formula for quality factor of a parallel RLC circuit.

$$Q = \omega_0 RC$$

Substitute 1000 for ω_0 , 100×10^3 for R and 10×10^{-6} for C .

$$\begin{aligned}Q &= \omega_0 RC \\ &= (1000)(100 \times 10^3)(10 \times 10^{-6}) \\ &= 1000\end{aligned}$$

Therefore, the quality factor, Q is 1000.

Step 3 of 6

Calculate the bandwidth.

$$\begin{aligned}B &= \frac{\omega_0}{Q} \\ &= \frac{1000}{1000} \\ &= 1\text{ rad/s}\end{aligned}$$

Therefore, the bandwidth of the circuit, B is 1 rad/s.

Step 4 of 6

If the resistor value is, $R = 200 \text{ k}\Omega$.

Calculate the resonant frequency.

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Substitute 100×10^{-3} for L and 10×10^{-6} for C .

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{LC}} \\ &= \frac{1}{\sqrt{100 \times 10^{-3} \times 10 \times 10^{-6}}} \\ &= 1000 \\ &= 1 \text{ krad/s}\end{aligned}$$

Therefore, the resonant frequency, ω_0 is 1 krad/s .

The resonant frequency does not depend on the value of resistor. Therefore, the value of resonant frequency does not change.

Step 5 of 6

Write the formula for quality factor of a parallel RLC circuit.

$$Q = \omega_0 RC$$

Substitute 1000 for ω_0 , 200×10^3 for R and 10×10^{-6} for C .

$$\begin{aligned}Q &= \omega_0 RC \\ &= (1000)(200 \times 10^3)(10 \times 10^{-6}) \\ &= 2000\end{aligned}$$

Therefore, the quality factor, Q is 2000 .

The quality factor is directly proportional to resistor value. Therefore, the value of quality factor is doubled.

Step 6 of 6

Calculate the bandwidth.

$$\begin{aligned}B &= \frac{\omega_0}{Q} \\ &= \frac{1000}{2000} \\ &= 0.5 \text{ rad/s}\end{aligned}$$

Therefore, the bandwidth of the circuit, B is 0.5 rad/s .

The value of the bandwidth is reduced.